

November Crossword

Kevin Chen, Wenhao Lu, Aaron Qin

1	2
3	

ONE ACROSS

Coach K has a set of basketball plays, each worth a positive integer of points less than or equal to 14, and he wants to design a set of plays so that no two different subsets of plays score the same total points. Let n be the largest total number of points he can achieve while following this rule. What is the sum of the digits of n ?

ONE DOWN

During a stretch of the NBA Cup, an analytics department is trying to find a relation between a certain team's PPG and their rank with the function $\text{PPG}(n) = an^2 + bn + c$, where n represents the rank of that team in the tournament.

The analysts have the following data:

- There are 30 teams participating in the NBA Cup.
- The bottom 10 teams (across all groups of the NBA Cup) have an average PPG of 92.1.
- The bottom 20 teams (across all groups) have an average PPG of 97.6.
- The last place team (across all groups) has a PPG of 90.

Let P be the highest PPG across all 30 teams. What is the sum of the digits of P ?

TWO DOWN

In the NBA Cup, LaMelo possesses the ability to teleport. He is setting himself up for an alley-oop, and so he tosses the ball in a straight line for $\frac{5}{3}$ meters, then teleports to that location. However, LeBron turns into an eagle and soars down, whacking the ball along the path of a straight line again, but this time for 2 meters. The ball then rolls for a path of $\frac{5}{3}$ meters and then the process keeps on repeating, in a way that analysts do not know. Eventually, analysts determine that the convex hull of LaMelo's teleportation positions forms an electromagnetic field in the shape of a tetrahedron whose four faces are all congruent triangles with side lengths $\frac{5}{3}, \frac{5}{3}, 2$. If the volume occupied by the electromagnetic field can be written in the form $\frac{a\sqrt{b}}{c}$, where $\gcd(a, c) = 1$ and b is not divisible by the square of any prime number, what is the value of $a + b + c$?

THREE ACROSS

Henrietta is livestreaming basketball dunks in hopes of becoming the next Stephen Curry. When a senior scout for the Lakers joins the livestream to consider drafting Henrietta for the NBA Cup, she decides to act cocky and show off her skills with an infamous three-pointer. The path of her last shot before ending the livestream can be modeled in the form $-\frac{1}{3}x^2 + ux + 3$, where $u = \tan t$ for some angle t . Consider the hoop to be the circle modeled by the equation $(x - 6)^2 + (y - 7)^2 = 1$. Suppose the ball grazes the hoop at a point $P = (x_p, y_p)$, leading to an airball and aura loss. Find the last two digits of $\lfloor 10000 \cdot x_p \rfloor$.

